

Juergen Gall

Fourier Transform and Image Pyramids

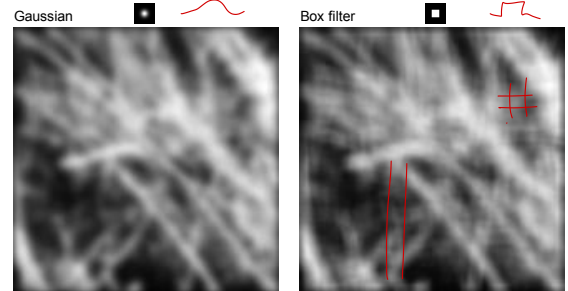
Computer Vision

MA-INF 2201 / MA-MOROB-M04
WS25/26

Linear filtering

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Gauss vs. box filter



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A sum of sines

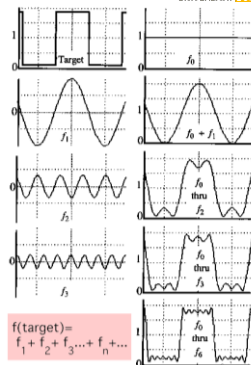
Our building block:

$$A \sin(\omega x + \phi)$$

A: Amplitude

 ω : angular frequency ϕ : phase

↳ Add enough of them to get any signal $f(x)$ you want!



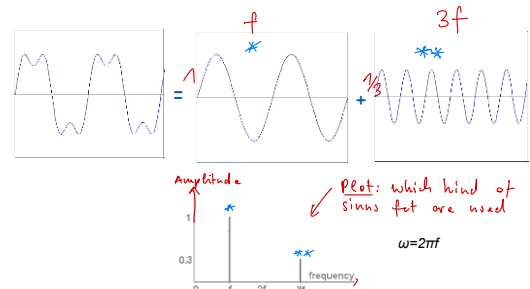
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Frequency Spectra

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$$\text{Example: } g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi(3f) t)$$



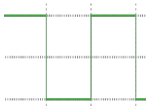
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Source: A. Efron

Frequency Spectra

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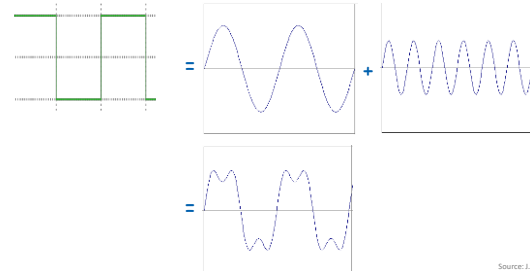
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Source: J. Hays

Frequency Spectra

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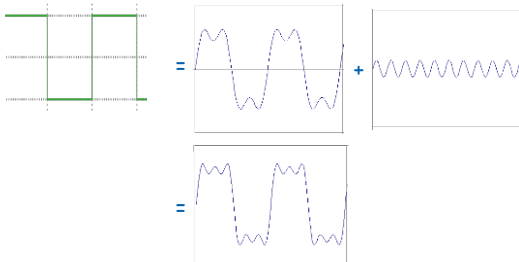


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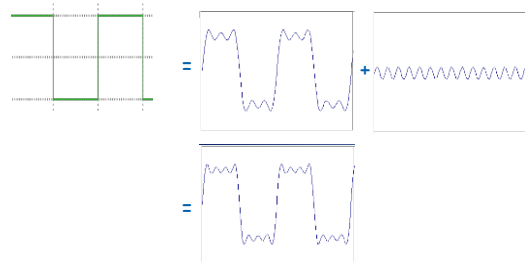
Source: J. Hays

Frequency Spectra



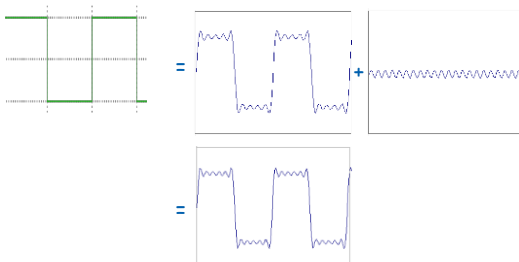
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Frequency Spectra



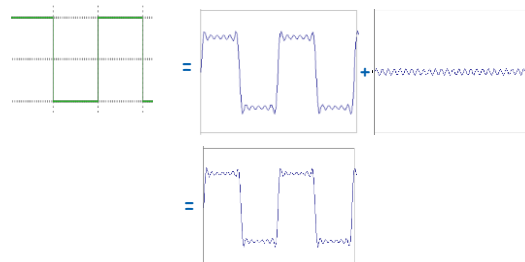
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Frequency Spectra



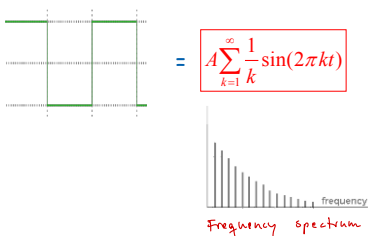
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Frequency Spectra



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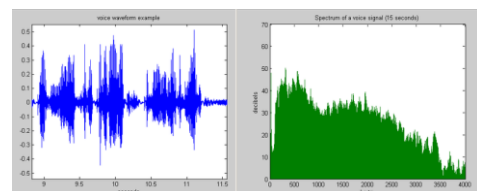
Frequency Spectra



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Example: Music

We think of music in terms of frequencies at different magnitudes



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Continuous Fourier transform

Euler's formula:

$$e^{\pm j\theta} = \cos(\theta) \pm j\sin(\theta)$$

Complex numbers:

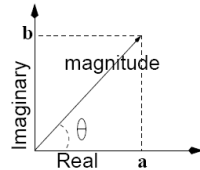
$$x = a + jb, \text{ where } j = \sqrt{-1}$$

$$\phi(x) = \tan^{-1}(b/a) \quad \text{phase}$$

$$|x| = \sqrt{a^2 + b^2} \quad \text{magnitude}$$

$$x^* = a - jb \quad \text{conjugate}$$

$$x = |x|e^{j\phi(x)}$$



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Continuous Fourier transform

1D Continuous Fourier Transform:

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du \quad \text{The inverse Fourier transform}$$

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx \quad \text{The Fourier transform}$$

2D Continuous Fourier Transform:

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi j(ux+vy)} dudv \quad \text{The inverse transform}$$

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi j(ux+vy)} dx dy \quad \text{The transform}$$

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Discrete Fourier transform

1D Discrete Fourier Transform:

$$f(x) = \sum_{u=0}^{N-1} F(u) e^{j2\pi ux/N} \quad \leftarrow \text{sometimes normalization factor also in inverse FT}$$

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-j2\pi ux/N}$$

2D Discrete Fourier Transform:

$$f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{M-1} F(u, v) e^{j2\pi j \left(\frac{ux}{N} + \frac{vy}{M} \right)}$$

$$F(u, v) = \frac{1}{N} \frac{1}{M} \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f(x, y) e^{-j2\pi j \left(\frac{ux}{N} + \frac{vy}{M} \right)}$$

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Fourier Transform

Fourier transform stores the magnitude and phase at each frequency

- Magnitude encodes how much signal there is at a particular frequency
- Phase encodes spatial information (indirectly)

$$\text{Amplitude: } A = \pm \sqrt{R(\omega)^2 + I(\omega)^2} \quad \text{Phase: } \phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$$

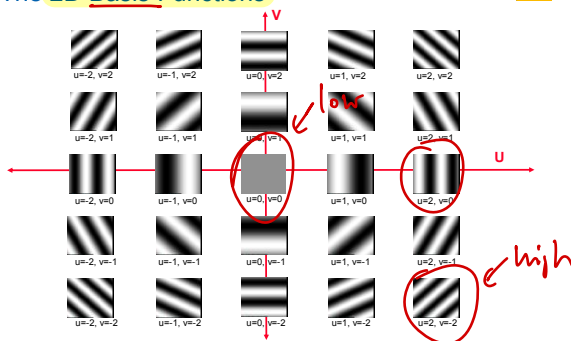
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Source: J. Hays

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The 2D Basis Functions



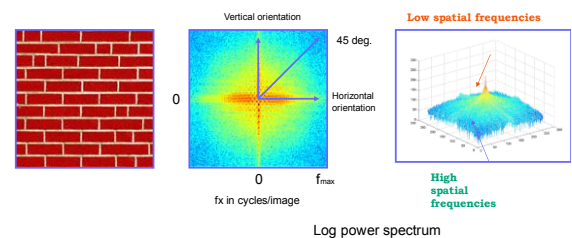
$\Rightarrow$ low frequencies in center,
high frequencies at borders

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How to interpret a Fourier Spectrum



Source: A. Tomrala

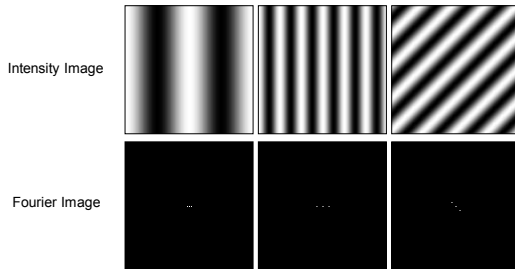
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Fourier analysis in images

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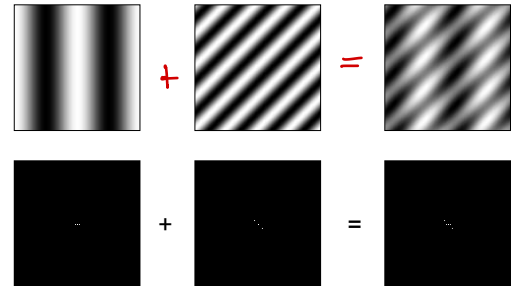


Source: J. Hays

Wieso 3 dots?

Signals can be composed

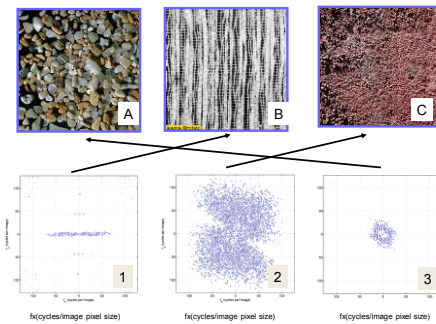
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Source: J. Hays

Fourier Amplitude Spectrum

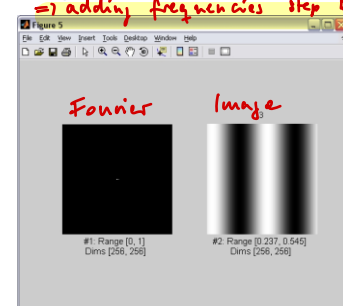
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more edges than A
⇒ higher frequencies

3 Fourier components

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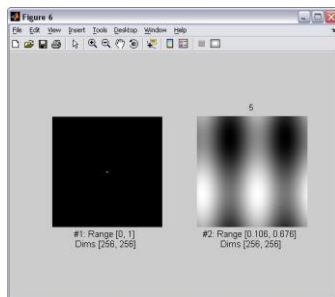


⇒ adding Descending order of magnitude.

Source: A. Torralba

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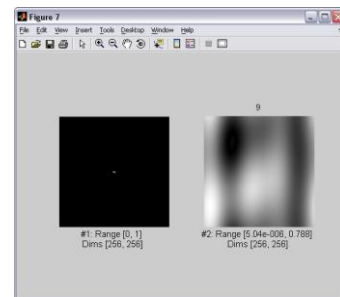


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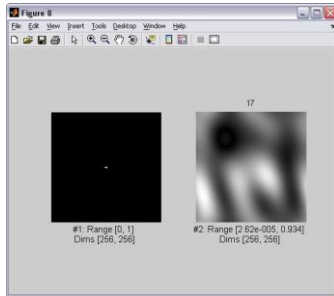
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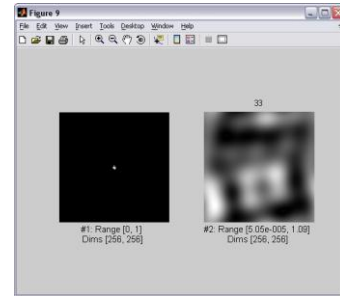


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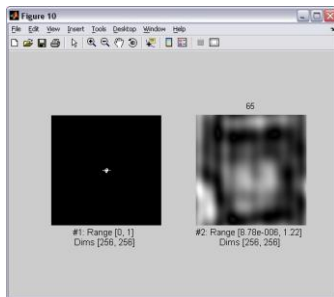


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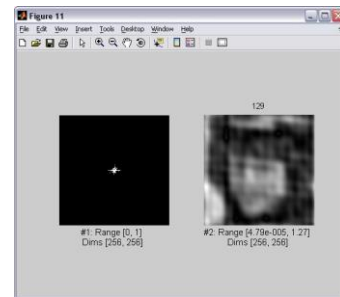


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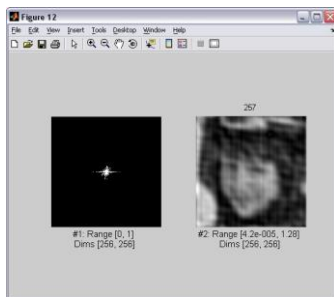


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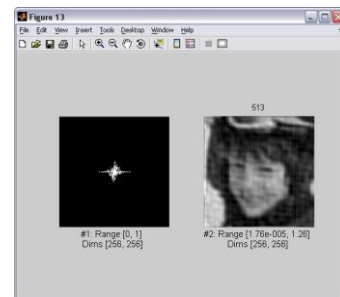


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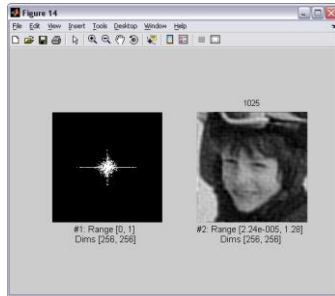


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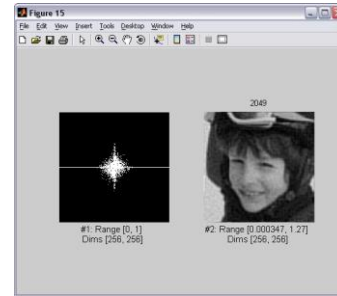


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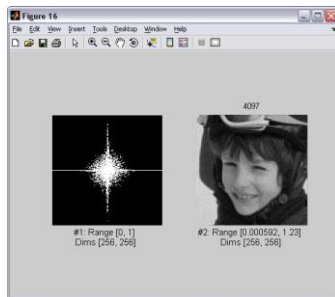


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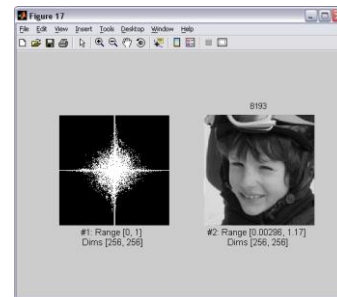


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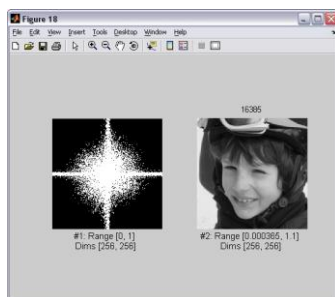


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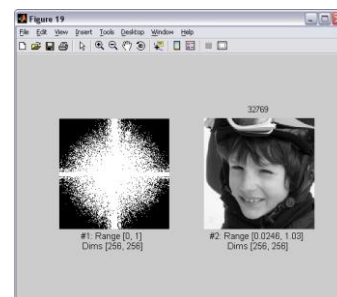


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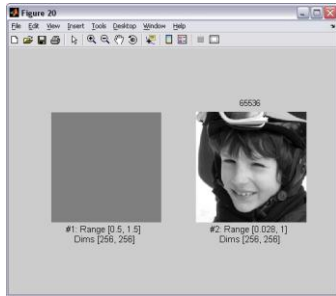
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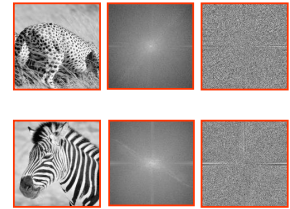
Fourier Transform

FREQUENCY
MAGNITUDESFREQUENCY
LOCATION

What frequencies
are present?
↑
Magnitude

Where are
the freq.
located
in image?
↑
Phase

- Fourier transform of a real function is complex
 - difficult to plot, visualize
 - instead, we can think of the phase and magnitude of the transform
- Phase is the phase of the complex transform
- Magnitude is the magnitude of the complex transform



Source: D. Forsyth

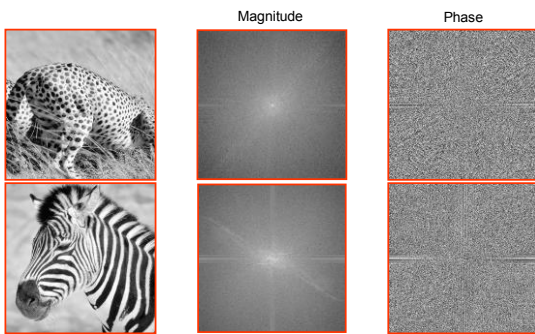
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Fourier Transform

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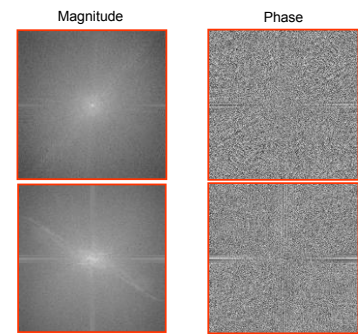
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Fourier Transform

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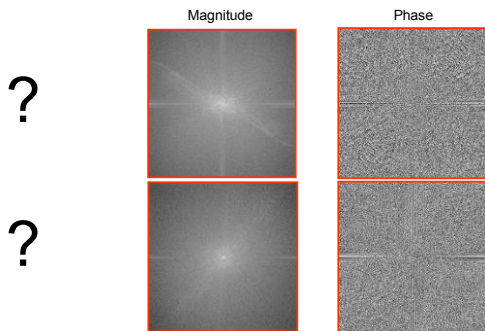
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Fourier Transform

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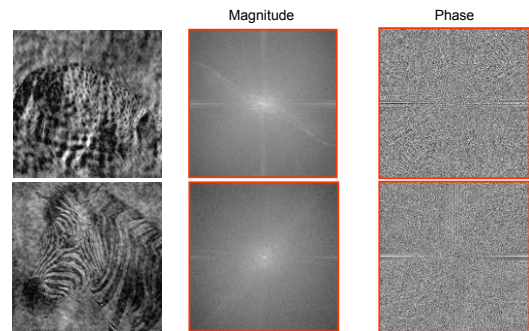
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Fourier Transform

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Shape preserved
 $\Rightarrow$ Location / Phase stays same

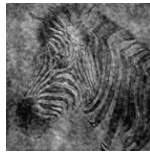
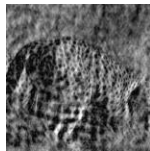
Phase and Magnitude



Image with cheetah phase
(and zebra magnitude)



Image with zebra phase
(and cheetah magnitude)



Source: D. Forsyth

The Convolution Theorem

- The Fourier transform of the convolution of two functions is the product of their Fourier transforms

$$F[g * h] = F[g] F[h]$$

convolution *product*

- The inverse Fourier transform of the product of two Fourier transforms is the convolution of the two inverse Fourier transforms

$$f[gh] = f[g] * f[h]$$

- Convolution in spatial domain is equivalent to multiplication in frequency domain!**

Source: J. Hays

Properties of Fourier Transforms

Property	Signal	Transform
superposition	$f_1(x) + f_2(x)$	$F_1(\omega) + F_2(\omega)$
shift	$f(x - x_0)$	$F(\omega)e^{-j\omega x_0}$
reversal	$f(-x)$	$F^*(\omega)$
convolution	$f(x) * h(x)$	$F(\omega)H(\omega)$
correlation	$f(x) \otimes h(x)$	$F(\omega)H^*(\omega)$
multiplication	$f(x)h(x)$	$F(\omega) * H(\omega)$
differentiation	$f'(x)$	$j\omega F(\omega)$
domain scaling	$f(ax)$	$1/a F(\omega/a)$
real images	$f(x) = f^*(x) \Leftrightarrow$	$F(\omega) = F(-\omega)$
Parseval's Theorem	$\sum_x [f(x)]^2 =$	$\sum_\omega [F(\omega)]^2$

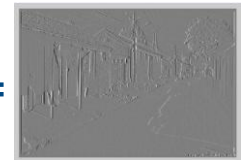
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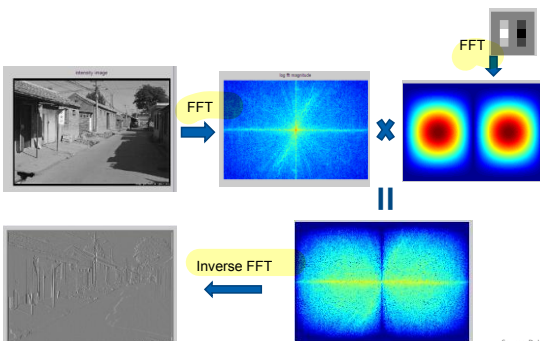
Filtering in spatial domain

1	0	-1
2	0	-2
1	0	-1



Source: J. Hays

Filtering in frequency domain



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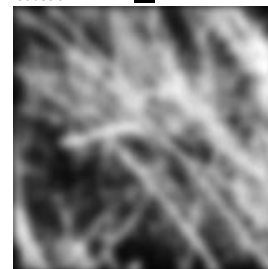
Source: D. Hofman

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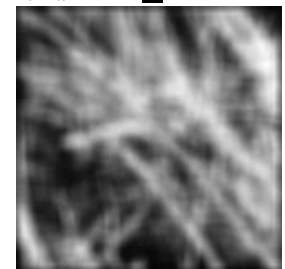
Linear filtering revisited

Gauss vs. box filter

Gaussian



Box filter



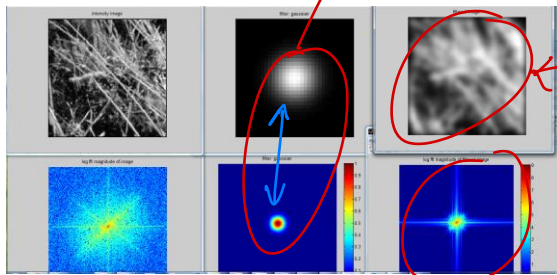
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$\Rightarrow$ could be faster to compute
 filtering in frequency domain!

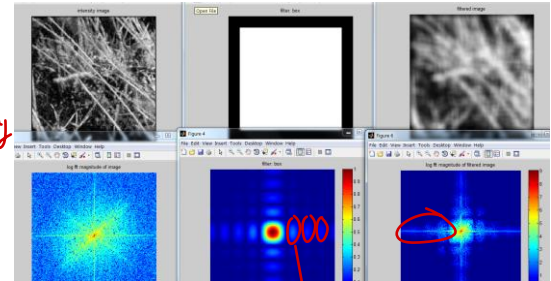
Gaussian



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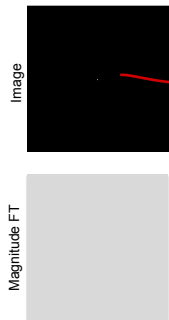
Gaussian removes high frequencies

Box filter



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Some important Fourier Transforms



20.10.2025 Jorgens Gell - Institute of Computer Science III - Computer Vision Group Source: A. Tomrala

delta fct:
 $\delta(x) = 0 \quad x \neq 0$
 $\int \delta(x) dx = 1$
 Fourier TF:
 $f(x) = \delta(x)$
 $F(u) = e^{-2\pi j u \cdot 0} = e^{-0} = 1 \leftarrow \text{constant case}$

Continuous Fourier transform

1D Continuous Fourier Transform:

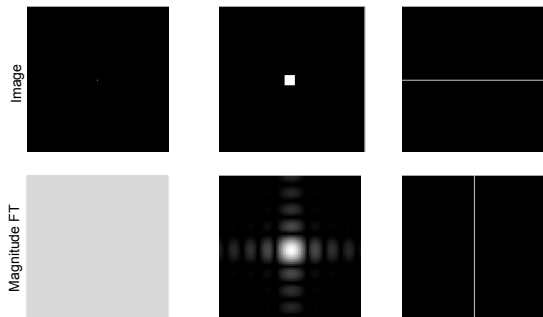
$$f(x) = \int_{-\infty}^{\infty} F(u) e^{+2\pi j u x} du \quad \text{The inverse Fourier transform}$$

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-2\pi j u x} dx \quad \text{The Fourier transform}$$

$$g(x) = \int \delta(x - u) h(u) du = h(x)$$

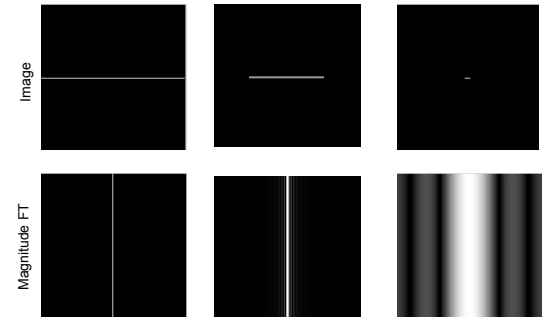
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Some important Fourier Transforms



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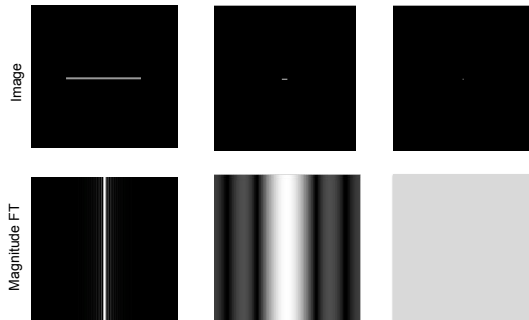
Some important Fourier Transforms



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Some important Fourier Transforms

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Source: A. Torralba

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Pairs - Functions

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Name	Signal	Transform
impulse	$\delta(x)$	$\Leftrightarrow 1$
shifted impulse	$\delta(x - a)$	$\Leftrightarrow e^{-j\omega a}$
box filter	$\text{box}(x/a)$	$\Leftrightarrow \text{sinc}(a\omega)$
tent	$\text{tent}(x/a)$	$\Leftrightarrow \text{sinc}^2(a\omega)$
Gaussian	$G(x; \sigma)$	$\Leftrightarrow \frac{\sqrt{2\pi}}{\sigma} G(\omega; \sigma^{-1})$
Laplacian of Gaussian	$(\frac{\partial^2}{\partial x^2} - \frac{1}{\sigma^2})G(x; \sigma)$	$\Leftrightarrow -\frac{\sqrt{2\pi}}{\sigma^3} \omega^2 G(\omega; \sigma^{-1})$
Gabor	$\cos(\omega_0 x)G(x; \sigma)$	$\Leftrightarrow \frac{\sqrt{2\pi}}{\sigma} G(\omega \pm \omega_0; \sigma^{-1})$

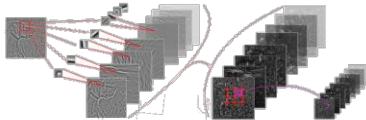
Handwritten notes:
 - Next to $\text{sinc}(a\omega)$: **artifacts of box filter**
 - Next to $\cos(\omega_0 x)G(x; \sigma)$: **complex in image domain**
 - Next to $G(\omega \pm \omega_0; \sigma^{-1})$: **simple in frequency domain**

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FFT and Convolutional Networks

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- Convolutional networks learn convolution kernels from training data

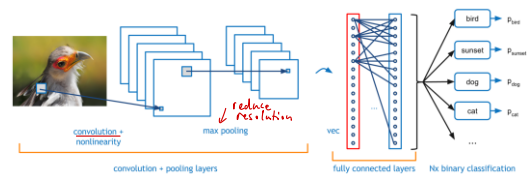


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FFT and Convolutional Networks

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- Convolutional networks learn convolution kernels from training data
- Can be used for image classification

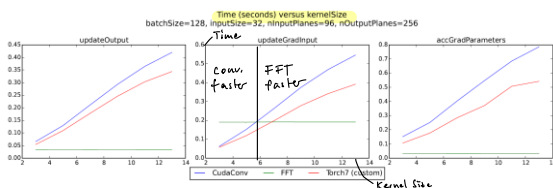


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FFT and Convolutional Networks

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- Convolutional networks are trained on GPUs
- Pixel-wise multiplications perfect for GPU
- NVIDIA CUDA Fast Fourier Transform library (cuFFT)
- Speed up training for larger kernels



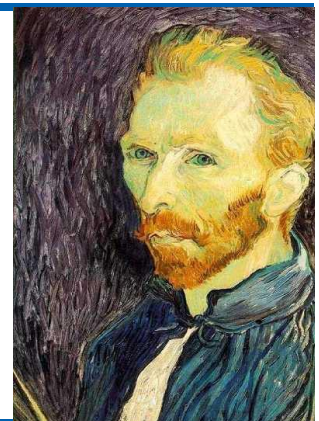
[Mathieu et al. Fast Training of Convolutional Networks through FFTs, ICLR2014]

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Image Scaling

This image is too big to fit on the screen. How can we reduce it?

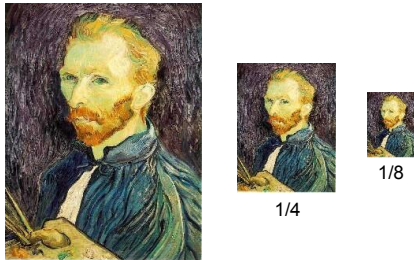
How to generate a half-sized version?



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Image Sub-Sampling

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Throw away every other row and
column to create a 1/2 size image
- called **image sub-sampling**

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Source: S. Narasimhan

Image Sub-Sampling

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1/2

1/4 (2x zoom)

1/8 (4x zoom)

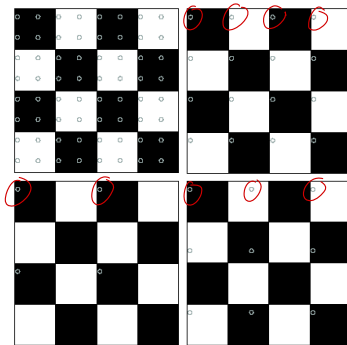
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Source: S. Narasimhan

Sampling

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Good sampling:
• Sample often or,
• Sample wisely

Bad sampling:

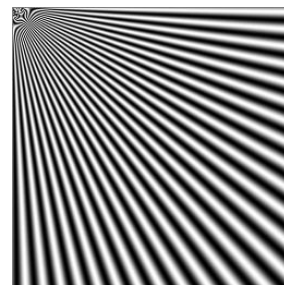
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Source: S. Narasimhan

Aliasing

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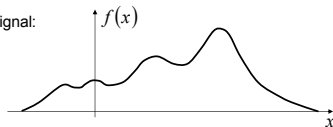
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Source: S. Narasimhan

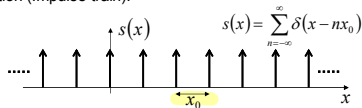
Sampling Theorem

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Continuous signal:



Shah function (Impulse train):



Sampled function:

$$f_s(x) = f(x)s(x) = f(x) \sum_{n=-\infty}^{\infty} \delta(x - nx_0)$$

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Source: S. Narasimhan

Sampling Theorem

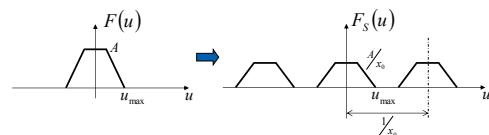
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Sampled function:

$$f_s(x) = f(x)s(x) = f(x) \sum_{n=-\infty}^{\infty} \delta(x - nx_0)$$

Sampling frequency $\frac{1}{x_0}$

$$F_s(u) = F(u) * S(u) = F(u) * \frac{1}{x_0} \sum_{n=-\infty}^{\infty} \delta\left(u - \frac{n}{x_0}\right)$$



Only if $u_{\max} \leq \frac{1}{2x_0}$

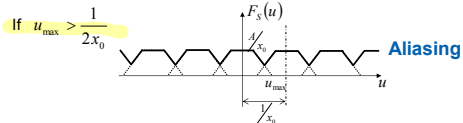
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Source: S. Narasimhan

Nyquist Theorem

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When can we recover $F(u)$ from $F_s(u)$?Only if $u_{\text{max}} \leq \frac{1}{2x_0}$ (Nyquist Frequency)

We can use

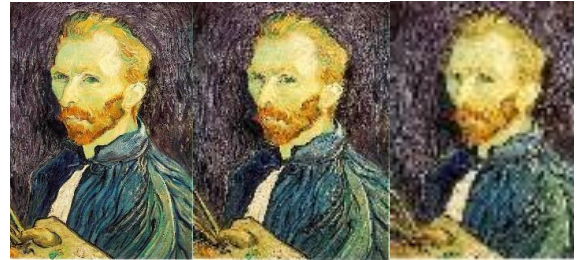
$$C(u) = \begin{cases} x_0 & |u| < \frac{1}{2x_0} \\ 0 & \text{otherwise} \end{cases}$$

Then $F(u) = F_s(u)C(u)$ and $f(x) = \text{IFT}[F(u)]$ Sampling frequency must be greater than $2u_{\text{max}}$

Source: S. Narasimhan

Image Sub-Sampling

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1/2

1/4 (2x zoom)

1/8 (4x zoom)

Source: S. Narasimhan

Sub-Sampling with Gaussian Pre-Filtering

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Gaussian 1/2

G 1/4

G 1/8

- Solution: filter the image, then subsample
- Filter size should double for each $\frac{1}{2}$ size reduction.

Source: S. Narasimhan

Sub-Sampling with Gaussian Pre-Filtering

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Gaussian 1/2

G 1/4

G 1/8

Source: S. Narasimhan

Compare with...

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1/2

1/4 (2x zoom)

1/8 (4x zoom)

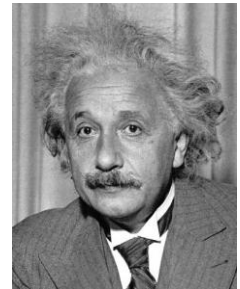
Source: S. Narasimhan

Template matching

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Goal: find  in image

Main challenge: What is a good similarity or distance measure between two patches?



Source: D. Hertz

Correlation

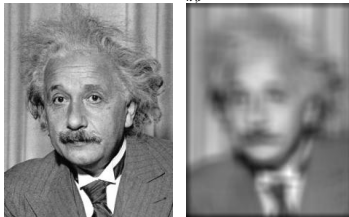
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Goal: find  in image

- Filter the image with eye patch (correlation)

$$h[m,n] = \sum_{k,l} g[k,l] f[m+k,n+l]$$

f = image
g = filter



What went wrong?

Input Filtered Image

Source: D. Hoiem

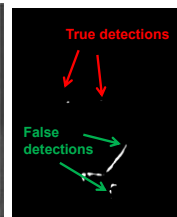
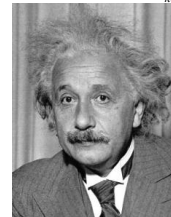
Zero-mean correlation

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Goal: find  in image

- Method 1: filter the image with zero-mean eye

$$h[m,n] = \sum_{k,l} (g[k,l] - \bar{g}) (f[m+k,n+l])$$



Input

Filtered Image (scaled)

Thresholded Image

Source: D. Hoiem

Sum Square Difference

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Goal: find  in image

- Method 2: SSD $h[m,n] = \sum_{k,l} (g[k,l] - f[m+k,n+l])^2$



Input

1- sqrt(SSD)

Thresholded Image

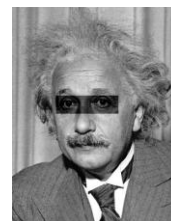
Source: D. Hoiem

Sum Square Difference

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Goal: find  in image

- Method 2: SSD $h[m,n] = \sum_{k,l} (g[k,l] - f[m+k,n+l])^2$



Input

1- sqrt(SSD)

Source: D. Hoiem

Normalized Cross Correlation

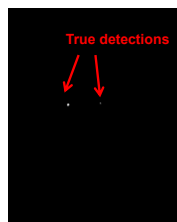
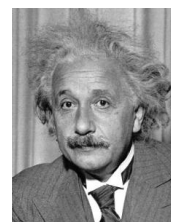
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Goal: find  in image

- Method 3: Normalized cross-correlation

$$h[m,n] = \frac{\sum_{k,l} (g[k,l] - \bar{g})(f[m+k,n+l] - \bar{f}_{m,n})}{\left(\sum_{k,l} (g[k,l] - \bar{g})^2 \sum_{k,l} (f[m+k,n+l] - \bar{f}_{m,n})^2 \right)^{0.5}}$$

mean template
mean image patch



Input

Normalized X-Correlation

Thresholded Image

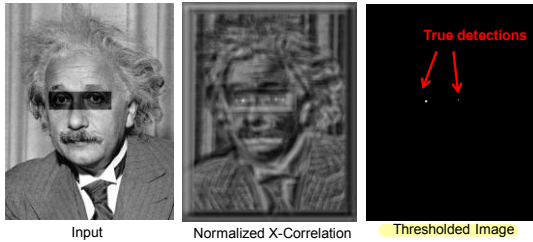
Source: D. Hoiem

Normalized Cross Correlation



Goal: find  in image

Method 3: Normalized cross-correlation



Input

Normalized X-Correlation

Thresholded Image

Source: D. Hoiem

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Fast template matching



Template

Search Region



How make it faster?

Source: S. Narasimhan

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Multi-resolution



Template

Search Region



Original Image



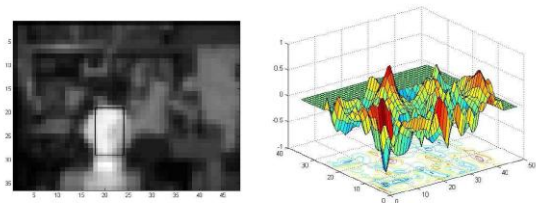
Source: S. Narasimhan

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Level 3 Search

- At the lowest pyramid level, we search the entire image with the correlation template



Source: S. Narasimhan

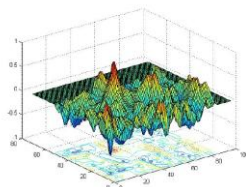
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Level 2 Search



- Subsequent searches are constrained to a neighborhood of only several pixels in the x and y directions



Source: S. Narasimhan

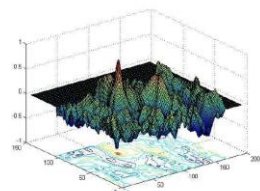
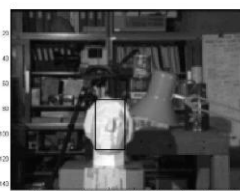
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Level 1 Search



- Subsequent searches are constrained to a neighborhood of only several pixels in the x and y directions



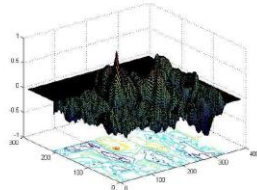
Source: S. Narasimhan

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Level 0 Search

- In the end, the total time (in Matlab) was reduced from ≈ 31 seconds to ≈ 0.5 seconds while obtaining the same template match



Source: S. Narasimhan

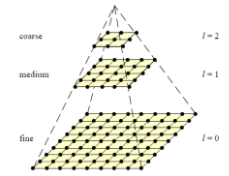
Coarse-to-fine Image Registration

Compute Gaussian pyramid

Align with coarse layer

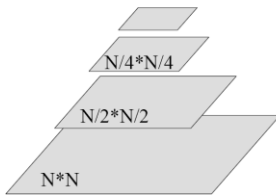
Successively align with finer layers

Search smaller range



Source: D. Holm

Space requirements

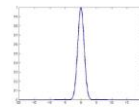


$$N^2 + \frac{1}{4}N^2 + \frac{1}{16}N^2 + \dots = 1\frac{1}{3}N^2$$

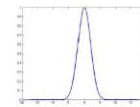
Source: S. Narasimhan

The Gaussian Pyramid

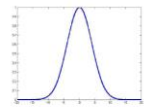
- Smooth with Gaussians because
 - a Gaussian * Gaussian = another Gaussian
- Synthesis
 - smooth and downsample
- Gaussians are low pass filters, so repetition is redundant
- Kernel width doubles with each level



Level 1



Level 2

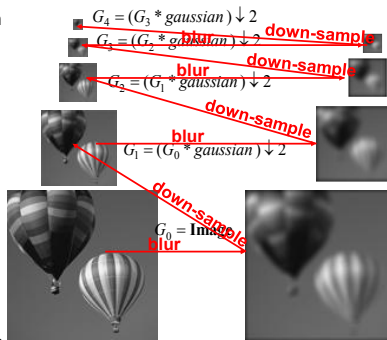


Level 3

Source: S. Narasimhan

Gaussian pyramid

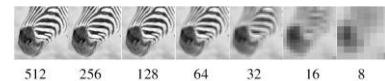
Low resolution



High resolution

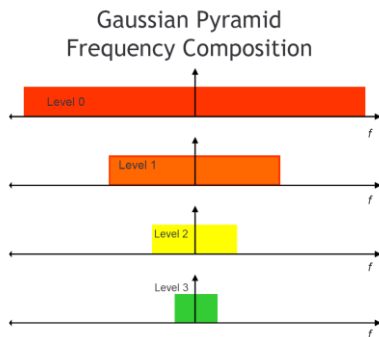
Source: S. Narasimhan

Gaussian pyramid



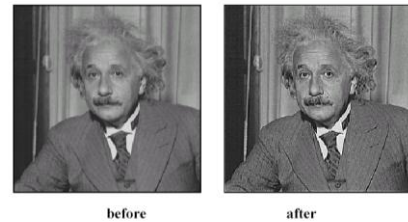
Source: D. Forsyth

Frequencies



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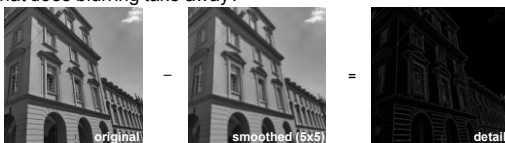
Sharpening revisited



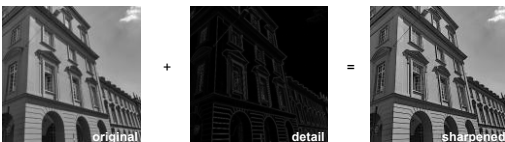
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Sharpening

What does blurring take away?

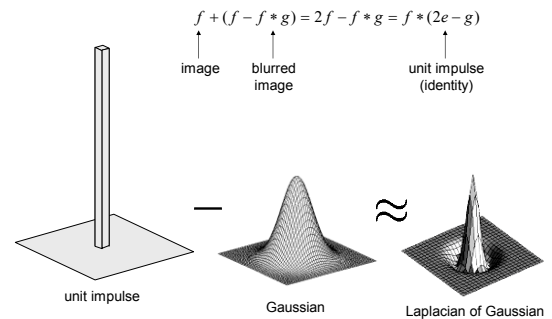


Let's add it back:



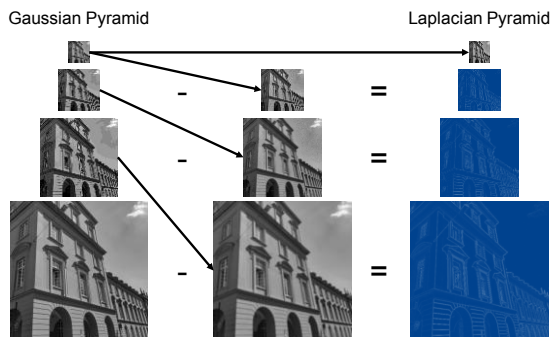
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Laplacian of Gaussian



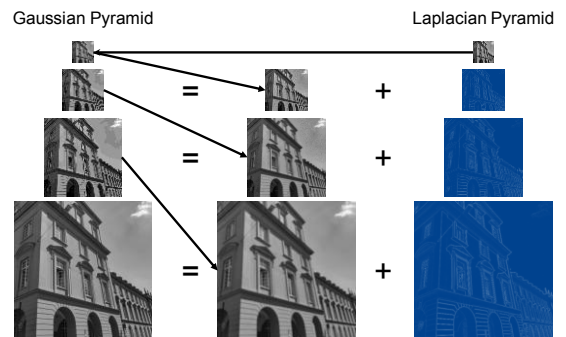
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Laplacian pyramid



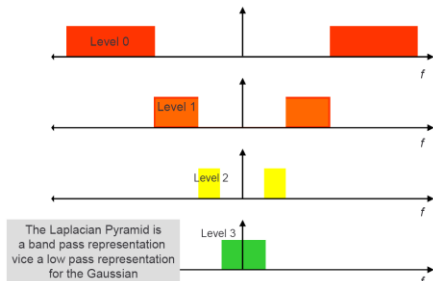
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Laplacian pyramid



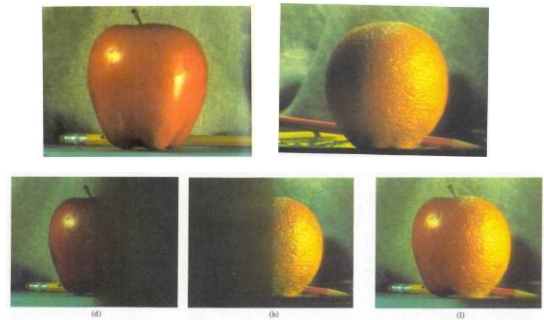
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Frequencies

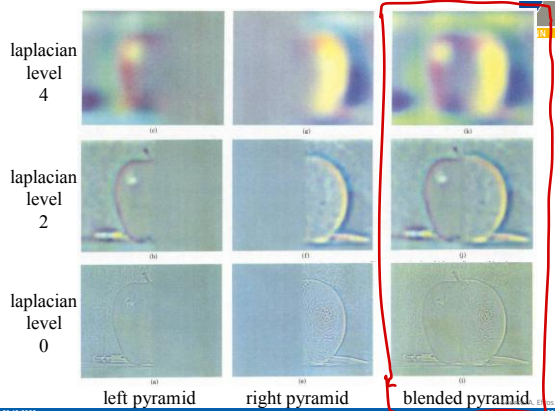
Laplacian Pyramid
Frequency Composition

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Pyramid Blending



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Blending Regions



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Laplacian Pyramid: Blending

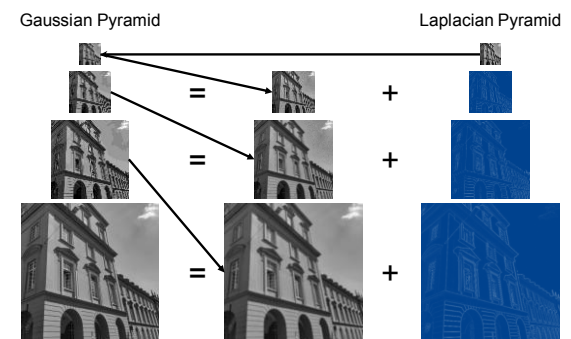
General Approach:

1. Build Laplacian pyramids LA and LB from images A and B
2. Build a Gaussian pyramid GR from selected region R
3. Form a combined pyramid LS from LA and LB using GR as weights:
 - $LS_i(i,j) = GR_i(i,j) * LA_i(i,j) + (1 - GR_i(i,j)) * LB_i(i,j)$
4. Collapse the LS pyramid to get the final blended image ($LS_f = LS_f + \text{expand}(LS_{f+1})$)

[P. Burt and E. Adelson. A multiresolution spline with application to image mosaics. ACM Trans. Graph. 1983]

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Laplacian pyramid



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Blending Regions



Source: A. Effros

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